

Lab Data Management API

Role: Backend Developer

Difficulty: junior

Project Type: backend

Duration: 3 days

Focus: fullstack

Candidate: Nikita Rane

Project Overview

Create a REST API for recording microscopy data with a simple React UI for CRUD operations.

Technologies

- React
- Node.js
- Express.js
- JavaScript
- MongoDB

Requirements

- Build a practical project
- Measurable and matches skills
- Introduce new learning
- Evaluate implementation and problem solving

Objectives

- Design API schema
- Implement CRUD endpoints
- Persist data in MongoDB
- Create React UI
- Test endpoints

Functional Requirements

- Create lab session record
- Add microscopy image metadata
- Retrieve list of sessions
- Update session details
- Delete session
- Search by date

Non-Functional Requirements

- Use JWT for auth
- Validate input
- Return JSON
- Handle errors gracefully
- Document API

Modules

Auth

JWT based authentication

- Login
- Register
- Token refresh

LabSession

CRUD for lab sessions

- Create
- Read
- Update
- Delete

ImageMetadata

Manage image details

- Add image
- List images
- Delete image

API

Express routes and controllers

- Route setup
- Middleware

UI

React front end

- Session list
- Session form
- Image upload

Implementation Plan

Setup

- Initialize repo
- Install dependencies
- Configure MongoDB

Schema Design

- Define User, Session, Image schemas

Auth

- Implement register/login endpoints
- Add JWT middleware

LabSession CRUD

- Create session endpoint
- Read session endpoint
- Update session endpoint
- Delete session endpoint

ImageMetadata CRUD

- Add image endpoint
- List images endpoint
- Delete image endpoint

UI & Testing

- Build React components
- Write unit tests
- Run Postman collection

Evaluation Criteria

API Design (25%)

Clear, RESTful endpoints

Code Quality (20%)

Clean, modular code

Error Handling (15%)

Graceful error responses

Testing Coverage (20%)

Unit & integration tests

Documentation (20%)

API and README

Deliverables

- Git repo
- API docs
- React app
- Postman collection
- Test suite
- README

Suggested Folder Structure

- src
- src/controllers
- src/models
- src/routes
- src/middleware
- client

API Endpoints

GET /sessions

List sessions

POST /sessions

Create session

GET /sessions/:id

Get session

PUT /sessions/:id

Update session

DELETE /sessions/:id

Delete session

POST /sessions/:id/images

Add image

GET /sessions/:id/images

List images

GET /sessions/search

Search by date

Database Design

users

Auth users

- username
- passwordHash
- role

lab_sessions

Session data

- title
- description
- date
- createdBy

images

Image metadata

- sessionId
- url
- description
- capturedAt

Testing Plan

- Unit tests for controllers
- Integration tests for routes
- API contract tests
- Load test
- Security test
- Manual UI test

Candidate Fit

Matches lab data recording skills, introduces backend and API design.